

L2
Introduction to [lang]  Java

Join Discord - <https://tinyurl.com/ly-discord>

Writing our first program

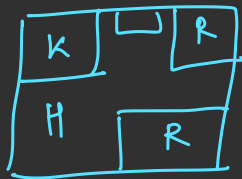
Java it is a OOPs language.

Blueprint

↳ Classes

↳ Objects

Analyze



↳ Key using which we access the class

Functionalities

Rule 1

↳ Java program should have class.

↳ First letter should be

Capital.

✓
Blueprint of House ~ Class

Main

Main → 99% ← Public vs Private only you can use
public property + anyone can use

File name → Main . java

```
public class Main  
{
```

Rule →

Filename should
be same as
Class name

Pathan

.txt

.pdf

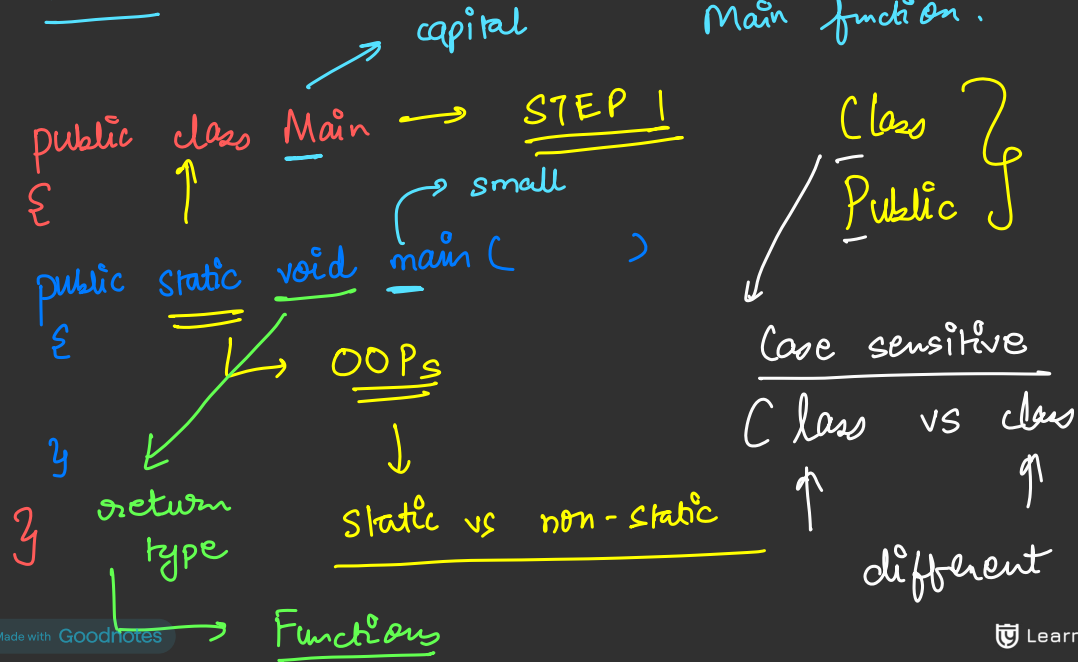
.png

.mp4

→ .java

3

Rule 3 → Execution of program starts from Main function.



Keyword → Fixed meaning
↳ Reserved

Saksham.java → Saksham

```
public class Saksham
{
    public static void main(String args[])
    {
    }
}
```

Array

org / org

String

↳ Detailed cover is String

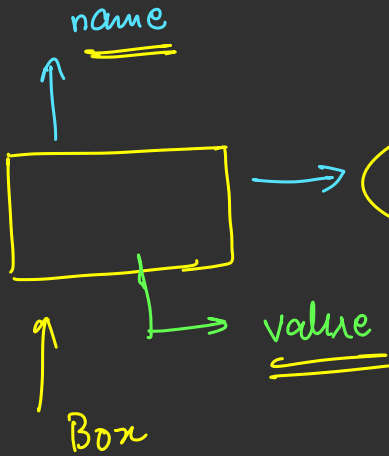
Understanding Class and our first program

```
public class Main
{
    public static void main (String
    {
        System.out.println
        ("Hello");
    }
}
```

Annotations and Concepts:

- public class Main**: **public** (green), **class** (yellow), **Main** (blue)
- Annotations**: **public** (green), **static** (green), **void** (green)
- Annotations**: **String** (green), **arg[]** (green)
- Annotations**: **System.out.println** (red), **("Hello")** (red)
- Annotations**: **File name** (red), **Main.java** (red)
- Annotations**: **Arrays** (red)

Variables → Store anything



type

10¹⁵

int

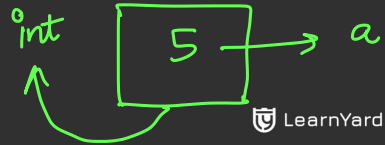
data type

type name = value ;

↓ ↓
int a = 5 ;

int → -2^*10^9 to 2^*10^9

long → -2^*10^{18} to 2^*10^{18}



Data types

`int` → store integer → $\frac{-2^{*}10^9 \text{ to } 2^{*}10^9}{(\text{approx})}$

`long` → store big value → $\frac{-2^{*}10^{18} \text{ to } 2^{*}10^{18}}{}$

`char` → 'a', 'b', 'A', '@', '?' (single letter)

↳ single (' ') (Quote)

`String` → English sentence / English word
↳ Multiple letters → " " (Double quote)

float $\longrightarrow$ Decimal (upto 8 places of decimal)

double $\longrightarrow$ Decimal (upto 15 places of decimal)
after decimal more digits

double

$$\left. \begin{array}{l} 7/4 = 1 \\ 7\%4 = 3 \end{array} \right\}$$

$$\begin{array}{r} 4 \overline{) 7} 1 \\ \underline{-4} \\ 3 \\ \underline{} \end{array}$$

Operators

Arithmetic Operator & Assignment Operator

$+$ $\rightarrow$ add

$-$ $\rightarrow$ subtract

$*$ $\rightarrow$ multiply

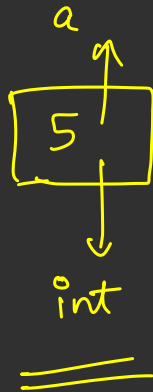
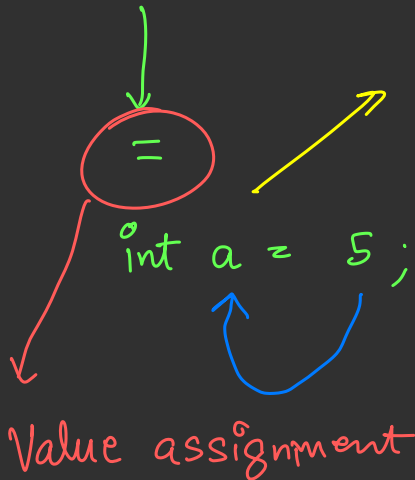
$/$ $\rightarrow$ divide

$\%$ $\rightarrow$ Remainder

$=$

`int a = 5;`

Value assignment



Operators

Relational and Logical Operator

↳ Conditional
Class

Taking Input

In Java - we use Scanner class

Input

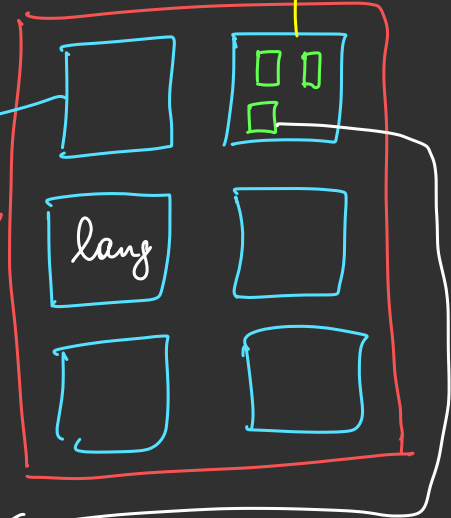
Package

group of
classes

Java

Scanner

util



I import java.util.Scanner;

Scanner
class
imported

Create object of Scanner

II → Scanner obj = new Scanner

Classname objectname = new Classname();

class

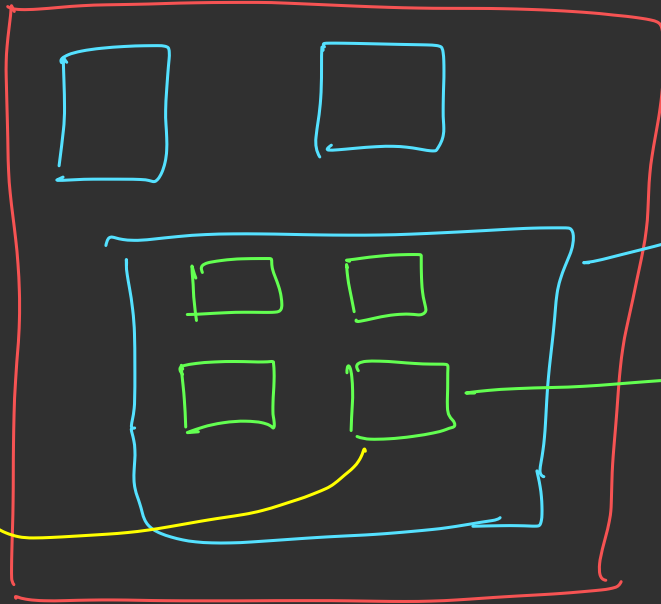
Functionality

Key → Object

new Int()

Create
object

nextInt()



util

Scanner

obj. nextInt()



Rule

String

object Name . functionality Name ()

newInt () → Input integer → 5

newFloat () → Input float number → 10.57

newLong () → Input long number → 10^{15}

new () → Input Word (single word)

newLine () → Input Sentence (multiple words)

Homework Question

1. Enter two numbers and perform all arithmetic operations
2. Enter P, R, T and calculate simple interest.

Next Class Plan

1. Conditional Statements
2. Loops
3. Some Practice

Thank You!

Reminder: Going to the gym & observing the trainer work out can help you know the right technique, but you'll muscle up only if you lift some weights yourself.

So, PRACTICE, PRACTICE, PRACTICE!